

2(a)(i)	<u>bone marrow</u> ;	1
2(a)(ii)	<i>accept points from a diagram</i> <i>max three from:</i> 1 detection / recognition ; e.g. detects (named type of) pathogen recognises, (foreign) antigens / antibodies complexed to antigens has receptors (for antigens) 2 engulfs / envelops, pathogen / bacterium / AW ; A phagocytosis occurs A endocytosis occurs A pseudopodia form 3 forms, vacuole / vesicle / phagosome ; 4 ref. to lysosome fusion ; 5 ref. to hydrolytic / digestive, enzymes ; A named examples A hydrolases 6 ref. to antigen presentation ; 7 AVP ; e.g. (response is) non-specific / innate	3
2(a)(iii)	<i>any one valid suggestion from:</i> produces inhibitors for / deactivates, lysosomal enzymes ; escapes out of phagosome ; forms resistant spore / is resistant to digestive enzymes ; AVP ; e.g. suggestion of macrophage malfunction	1

Question	Answer	Marks
2(b)	any three from: 1 production of mucus by, mucous glands / goblet cells ; 2 sticky / AW, mucus or 3 mucus traps, pathogens / bacteria / microorganisms ; 4 mucus acts as a barrier (to prevent entry) ; 5 mucus increases distance to reach cells ; 6 cilia on ciliated epithelial cells ; A ciliated epithelium 7 cilia, waft / move, mucus / AW ; 7 <i>idea that</i> (contaminated) mucus is moved, away from alveoli / away from lung tissue / towards back of mouth / AW ;	3
2(c)	any one suggestion from: blood / plasma / circulatory system ; lymph / lymph system ; within, neutrophils / macrophages / phagocytes ; A white blood cells / leucocytes	1
2(d)	any one from: from infected, cows / cattle ; eating contaminated, meat / beef (from infected cattle) ; drinking, raw / unpasteurised, milk (from infected cows) ;	1
2(e)	any two from: increases chances of, killing / AW, all bacteria ; I will kill all bacteria (unqualified) A will kill all bacteria if bacteria are, susceptible / not resistant if bacteria are resistant to one antibiotic, then still susceptible to other antibiotics / AW ; reduces chance of mutations arising / (if bacteria susceptible) mutations unlikely to occur against all antibiotics ; AVP ; e.g. antibiotic can be stopped because of side effects idea that more effective because different antibiotics will work on different targets (in the bacteria)	2

Question	Answer	Marks
2(f)	any two from: overall trend for both is increase ; rate of increase in cases of MDR-TB greater than rate of increase in cases of TB / AW / ora ; ref. to fluctuations / detail of fluctuations / periods of decrease ;	2
2(g)	any three from: <i>overall increase in TB cases:</i> 1 ref. to problems with people living in, poorly ventilated accommodation / close proximity to each other / AW ; 2 vaccination programmes not able to prevent increase / herd immunity insufficient / AW ; <i>decrease / numbers fairly constant:</i> 3 ref. to (some) success of, prevention / treatment programme ; A described examples <i>increase in MDR-TB:</i> 4 people fail to finish course of treatment for normal TB / AW ; 5 ref. to ease of transmission of TB means easy to transmit <i>MDR-TB</i> ; <i>decrease in 2014 for MDR-TB:</i> 6 suggestion for decrease ; e.g. better, surveillance / cooperation, in completion of drug therapy more successful antibiotics used people better educated about preventing spread <i>for either increase in TB cases or increase in MDR-TB:</i> 7 qualified ref. to link between, HIV / HIV/AIDS, and TB ; 8 less money available for, longer drug treatment / vaccination, by, health authorities / governments ; 9 AVP ; e.g. balance between new cases and successfully treated cases means problem still exists in prevention ref. to changes in population size ref. to improvements in data collection ref. to immigration	3